

Modeling Task Classes and Service Time on a Server

The service time of a task depends on its computational size and on the processing speed of the server.

1. TASK CLASSES

Each task class k is characterized by a computational requirement C_k (e.g., CPU cycles, operations).

Computational size
of the task

$$C_k$$

Example:



Class A (delay-sensitive)

Lightweight tasks
(e.g., Fire detection)

$$C_A \sim \text{Exp}(\bar{C}_A)$$

mean = $\bar{C}_A$



Class B (delay-tolerant)

Heavy tasks
(e.g., biodiversity analysis)

$$C_B \sim \text{Exp}(\bar{C}_B)$$

mean = $\bar{C}_B$

Different classes have different average computational sizes $\bar{C}_k$.

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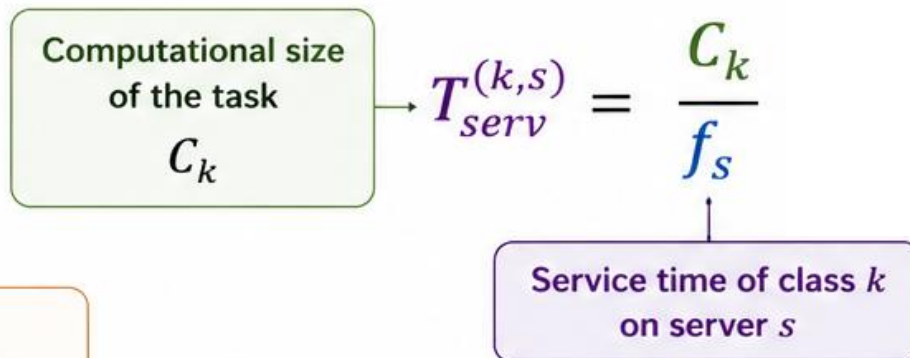
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If C_k is exponentially distributed, then

$T_{serv}^{(k,s)} = \frac{C_k}{f_s}$ is also exponentially distributed

with mean $\mathbb{E}[T_{serv}^{(k,s)}] = \frac{\bar{C}_k}{f_s}$.

The service rate is $\mu_{k,s} = \frac{1}{\mathbb{E}[T_{serv}^{(k,s)}]} = \frac{f_s}{\bar{C}_k}$.

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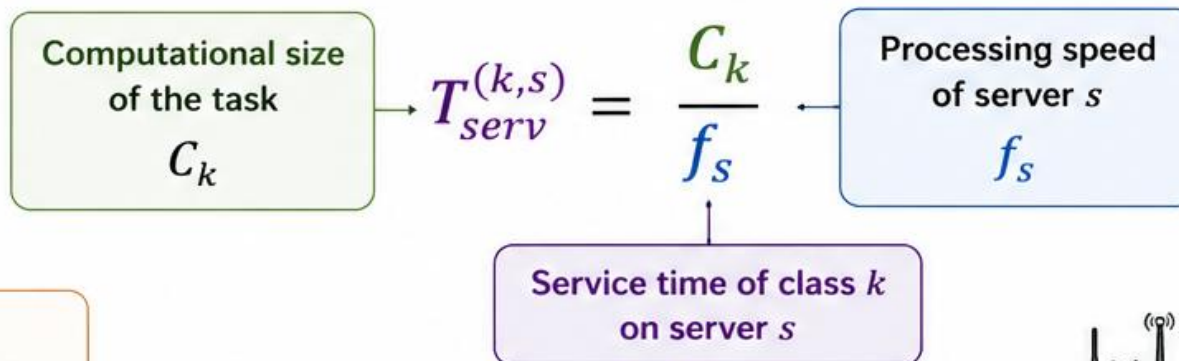
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2. SERVERS

Each server s is characterized by its processing speed f_s (operations/s).

Example:



Sensing Node (Local)

Limited capabilities f_{loc}
(e.g., 10^8 op/s)



HAPS Edge Server

High capabilities f_{haps}
(e.g., 10^9 op/s)

A faster server (higher f_s) provides smaller service time.

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3. NUMERICAL EXAMPLE

Task Class	Mean computational size $\bar{C}_k$ (cycles)	Deadline D_k^{max}
Class A (delay-sensitive)	5×10^6	0.5 s
Class B (delay-tolerant)	5×10^7	-



Mean service time $E[T_{serv}^{(k,s)}] = \frac{\bar{C}_k}{f_s}$			
Server	Processing speed f_s (cycles/s)	Class A (mean service time)	Class B (mean service time)
 Local (sensing node)	10^8	$\frac{5 \times 10^6}{10^8} = 0.05 \text{ s}$	$\frac{5 \times 10^7}{10^8} = 0.50 \text{ s}$
 HAPS Edge Server	10^9	$\frac{5 \times 10^6}{10^9} = 0.005 \text{ s}$	$\frac{5 \times 10^7}{10^9} = 0.05 \text{ s}$



Key takeaway: Service time decreases when the task is smaller (smaller $\bar{C}_k$) or the server is faster (larger f_s).
Different task classes and different server capabilities create heterogeneous service times and QoS performance.